

A Protocol for Human-Centric Adaptive User Interfaces: From Static Interaction to Behaviour-Driven Adaptation

Davide Matteri

*Department of Innovative Technologies
SUPSI*

Viganello, Switzerland

<https://orcid.org/0000-0002-2142-9776>

Sara Masiero

*Department of Innovative Technologies
SUPSI*

Viganello, Switzerland

<https://orcid.org/0000-0001-7158-9172>

Samuele Dell'Oca

*Department of Innovative Technologies
SUPSI*

Viganello, Switzerland

<https://orcid.org/0000-0002-6773-9155>

Elias Montini

*Department of Innovative Technologies
SUPSI*

Viganello, Switzerland

<https://orcid.org/0000-0002-6176-6005>

Vincenzo Cutrona

*Department of Innovative Technologies
SUPSI*

Viganello, Switzerland

<https://orcid.org/0000-0001-7830-7596>

Luca Canetta

*Department of Innovative Technologies
SUPSI*

Viganello, Switzerland

<https://orcid.org/0000-0002-3207-7470>

Abstract—Manufacturing production systems widely employ Human Machine Interfaces to guide workers and enable interactions with machines. The design of such interfaces often prioritize technological capabilities over User Experience design principles, resulting in standardized interfaces that do not adequately address user needs and provide them with excessive information. Adaptive User Interfaces present significant opportunities for enhancing traditional interfaces and improving the interactions with automation systems. In this work, a novel protocol is proposed to guide the design of adaptive, user-centered interfaces tailored to manufacturing contexts, shifting traditional static User Interfaces to advanced interfaces capable of adapting to users' behavior and characteristics. The protocol was applied in a controlled laboratory environment simulating an industrial assembly task, and 40 participants completed assembly tasks with the support of static and adaptive interfaces. The results demonstrated the protocol's effectiveness, highlighting that adaptive interfaces significantly reduce assembly time variability and errors, while enhancing user satisfaction by personalizing interactions based on user behavior.

Index Terms—Adaptive User Interface, Industry 5.0, Human-Machine Interaction, User Experience, Human-Centered Design

I. INTRODUCTION

The advent of Industry 5.0 (I5.0) is reshaping manufacturing by prioritizing human-centric goals such as worker well-being, safety, and productivity alongside technological advancements, especially Artificial Intelligence (AI) [1], [2]. A key challenge in this shift is balancing these objectives with efficiency and productivity, while improving Human Computer Interaction (HCI) [3], [4]. User Interfaces (UIs) represent a key element of I5.0, enabling communication interaction between users and systems by facilitating the execution of tasks. However, in high-pressure manufacturing environments, traditional UI often fail to accommodate varying skill levels, task complexities,

and real-time operational needs, relying mainly on context features and enabling technologies, such as hardware [5].

In this context, Adaptive User Interfaces (AUIs) represent a key factor in helping operators interact with increasingly complex systems, also boosting efficiency, promoting learning and enhancing the overall User Experience (UX). AUIs enable the dynamic adaptation of interfaces, tailoring them to user needs and preferences, also considering contextual factors such as skill level, task complexity, and environmental variables. This approach aligns with research on human-centered adaptive Human Machine Interfaces (HMIs) [6], context-aware Mixed Reality (MR) interfaces [7], and pattern-based adaptive interface supported by behavioral models [8]. For instance, context-aware AUIs have shown significant benefits in a manufacturing environment, improving productivity and reducing errors [9], providing task-specific guidance through various modalities, enabling non-specialists to perform complex tasks with minimal training. Adaptive multi-modal UIs facilitate real-time interaction with physical objects [10] and dynamically adjust based on contextual parameters through predefined adaptation rules.

While many forms of dynamic adaptation are known (e.g., adjusting information density, modifying control layouts, providing task-specific assistance, or guiding users through complex procedures with minimal cognitive effort [11]–[13]) several challenges are still to be faced when realizing AUIs:

- Algorithmic adaptability remains constrained by the limitations of real-time contextual understanding [14]. Systems can dynamically adjust layouts or information density, but inferring user intent across diverse scenarios still requires significant advances in multi-modal sensing and predictive modeling [3], [15]. For instance, misinterpretations of operator behavior or environmental conditions

may lead to inappropriate adaptations, such as oversimplifying interfaces for expert users or overwhelming novices with excessive data [10].

- User trust in AUI remains a critical psychosocial hurdle [16]. Operators often exhibit skepticism toward automatic and opaque interface modifications, and may tend to disable adaptation. Transparent adaptation logics and explicit user feedback contribute to build user trust.
- Lack of frameworks supporting the AUI development life-cycle [17]. Frameworks and methodologies are under-investigated in literature. Systematic approaches are desirable, particularly in relation to UX design and usability.

The present work is organized as follows. Section II systematically reviews theories and research on AUIs. Section III introduces the protocol for designing AUIs, detailing its steps and implementation. Section IV presents its application in a simulated manufacturing scenario involving a LEGO® forklift assembly task. Finally, Section V summarizes findings and suggests future research directions.

II. RELATION TO EXISTING THEORIES AND WORK

To review recent studies on AUIs in manufacturing contexts, this work employed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). The scope of the review was restricted to studies published in English between January 2019 and January 2025. Scopus and Web of Science were selected as source databases, and articles were retrieved by using the following set of keywords: *adaptive interface*; *adaptive user interface*; *adaptive system*; *manufacturing*; *human computer interaction*; *human machine interface*; *human-centered*; *intelligent user interface*; *user experience*; *user centered design*. Keywords were combined with logical operators to collect a comprehensive list of articles ('OR'), or relate technological solutions with domain of interest ('AND'). The following criteria were used to filter relevant studies:

- Studies must examine HMI and HCI in industrial manufacturing contexts, with a focus on UIs.
- Studies must describe and analyze the use of both adaptive and non-adaptive UI systems.
- Studies must provide empirical evidence supporting their conclusions, including data obtained through evaluation methods, and usability studies.

An overview of the results is presented in Fig. 1. 354 studies were retrieved from the two databases (219 from Scopus, 135 from Web of Science). Starting from these, 269 studies were excluded by screening titles and abstracts. At this stage, the selection of studies was guided by their relevance to HMI and HCI in manufacturing contexts, with a particular focus on UI design and adaptivity mechanisms, ultimately leading to the selection of 16 articles. Table I provides an overview of the final studies selected, along with the year and topics that contributed to the study.

Over the past few years, several systems have been created and tested for diverse manufacturing applications. Not only production, but also supply chain management [32],

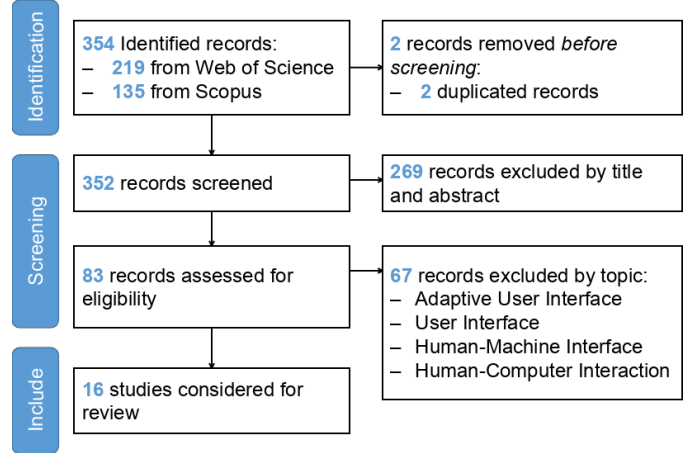


Fig. 1. Overview of the selection process with the PRISMA method

TABLE I
STUDIES SELECTED FROM LITERATURE REVIEW

Source	Year	Topic
[18]	2019	Designing 3D printer software interfaces for non-technical users
[19]	2021	AUI using nonlinear autoencoders to enhance human-machine interaction
[20]	2021	Human-centered framework for selecting worker assistance systems
[21]	2021	Design of models HMIs for software applications in I4.0
[22]	2022	Conceptual model of the Internet of Behaviors for adaptive intelligent systems
[11]	2022	Reference architecture and algorithms for adaptive assistance systems in industrial environments
[23]	2022	Interface and interaction design principles for MR
[24]	2023	Framework for designing adaptive dynamic automation between humans and machines
[25]	2023	Framework for designing AUI in XR environments
[26]	2023	Human-Machine Interaction dataset for AUIs
[27]	2024	Automated methods for evaluating usability and UX in smart products
[28]	2024	Adaptive virtual training system based on universal design
[29]	2024	Framework and development of a human-centered UI/UX model for a collaborative system
[30]	2024	User preference analysis for AI-assisted strategy recommendation systems
[31]	2024	Design guidelines for cognitive ergonomics in human-centered cobot applications
[15]	2024	Framework for the development of AUI in S-PSS

personnel skill development [33] and equipment maintenance [34]. The design and principles of HMI and HCI emerge as central themes in most studies, with a strong emphasis on personalization and UX evaluation for human-centered systems. Within the context of manufacturing systems, these concepts are frequently associated with worker support and training systems, including Extended Reality (XR), Virtual Reality (VR) and Augmented Reality (AR). These studies make significant contributions to the design of human-centered

UI, providing methodologies and tools for developing more intuitive, efficient, and adaptive solutions. For example, Reguera-Bakhache et al. proposes a data-driven methodology that uses Machine Learning (ML) to analyze operator interactions with industrial HMIs and generate temporal adaptation rules, optimizing interface usability in multi-operator scenarios [9]. Their approach led to over 40% reduction in operator interaction time and nearly 60% fewer clicks, significantly improving working experience. In contrast, Grandi et al. extend the scope of adaptivity by incorporating not only HMI functionalities but also operator skills and digital tools, systematically linking them to circular economy objectives, interaction modes, device types, and adaptive behaviors [35]. Building further on the goal of enhancing operator support, Keskin et al. propose a modular architecture that integrates Large Language Models (LLMs) into the HMI layer, allowing shop-floor personnel to query and visualize complex real-time sensor data using natural language [36]. Their system, tested in a simulated extrusion process scenario, successfully interprets and responds to user prompts through SQL and coding agents, demonstrating the potential of LLMs to bridge the gap between data complexity and operator expertise in dynamic manufacturing contexts. However, while these works offer valuable contributions to the engineering aspects of adaptive system development, they address only one phase of the broader process required to truly design AUIs, focusing predominantly on technical solutions or isolated validations [15]. Individuals and companies with limited programming expertise need accessible, step-by-step guidance to develop interfaces that are tailored to their specific operational needs. Moreover, there is a notable lack of standardization in the design of AUIs, which make difficult the creation of consistent and effective interaction paradigms. Most current solutions are tailored to single-user contexts, despite an increasing demand for systems capable of dynamically adapting to multiple users simultaneously. A key challenge lies in the absence of a clear and shared definition of the components that constitute UI adaptation [37].

By incorporating adaptive and user-centered features, personalized experiences can be created that enhance learning, support individual performance, and improve ergonomic assistance compared to current support systems [11]. This study explores these critical aspects by addressing the following research questions:

- What key factors should be considered in AUIs compared to traditional static UIs to ensure a user-centered design?
- What methodologies can be developed to dynamically adapt traditional UIs to user-centered characteristics in smart manufacturing systems, addressing their needs and enhancing usability, safety, and performance?

III. THE PROTOCOL FOR AUI DESIGN

The proposed protocol focuses on the design and development of user-centered adaptive logic for AUIs. The protocol offers a systematic approach that combines the principles outlined in the *morphological box for adaptation design of assistance systems* [11] and the five general dimensions of the

TABLE II
INTERFACE COMPONENTS AND ELEMENTS

Component	Elements
Task	Visualization, tracking, progression indicators
Machine	Status indicators, control panels
Instructions	Step-by-step guidance, multimedia support
Errors	Troubleshooting guidance, corrective action prompts
Feedback	Task performance, system response
Logging	Button clicks, menu selections, form completions, task completion sequences and timing, error occurrences, recovery attempts, system state changes

adaptive context life-cycle model [25]. The framework serves as a tool for developers and designers, providing guidance in the critical stages of AUI design, and it is made of seven steps, which are depicted in Fig. 2 and detailed in the following.

1) *Static interface design*: This step sets the foundation for identifying adaptation needs and relevant interaction features. It requires to design a static interface prototype with clear information architecture that aligns with the natural workflow. The key components of the interface and their associated elements are summarized in Table II.

2) *Static interface testing*: This step focuses on usability tests to understand users' interactions with the static interface, documenting performance metrics and user feedback. It requires recruiting participants representing the target user population, ensuring diversity in experience levels, ages, and technical backgrounds. It is necessary to prepare a testing environment that authentically simulates the industrial scenario with appropriate physical setup, lighting conditions, and ambient noise. During testing sessions, multiple data collection methods should be considered:

- Conduct pre-test interviews to gather baseline information about participants' experience and expectations.
- Employ think-aloud protocols during task execution where participants verbalize their thoughts while interacting with the interface.
- Track logs through the functionality developed in the static UI for collecting valuable insights about the user's interaction behavior.
- Conduct post-task semi-structured interviews focusing on the perceived usefulness of specific interface elements, e.g., understanding which information elements were most critical, and challenges encountered during task execution, moments of confusion or increased cognitive load, e.g., understanding at what points the user was feeling uncertain about the next steps.

3) *Interaction data and user feedback analysis*: Quantitative analysis of interaction logs must be performed to measure task completion times for all steps, error rates and recovery patterns, navigation efficiency (path length through interface elements), information access patterns (frequency and duration of consulting specific information) and idle periods suggesting confusion or decision-making challenges. After this analysis, it is necessary to perform sequential pattern mining on interaction logs to identify common behavioral sequences that

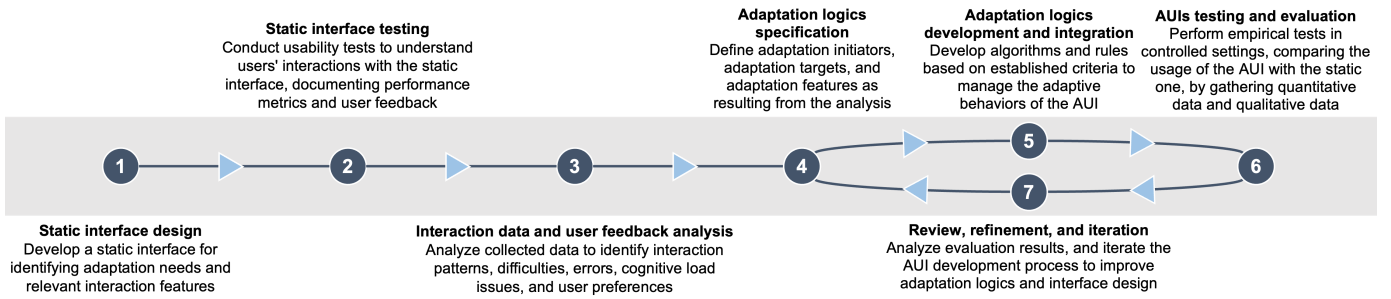


Fig. 2. The seven steps of the proposed protocol for AUI development

precede errors or successful task completions. Apply time-series clustering techniques to group similar interaction patterns and identify distinct user strategies. Construct a feature engineering framework that transforms raw interaction data into meaningful indicators of user states and needs. Finally, comparative analysis must be performed between user groups (e.g., novice vs. experienced) to identify differential needs and interaction patterns that suggest adaptation opportunities. Heat maps of interface element usage help visualizing attention distribution and identifying underutilized or overused features.

4) *Adaptation logics specification*: Use structured brainstorming techniques to identify adaptation scenarios based on the patterns uncovered in the previous analysis. Fig. 3 presents a framework that supports the design of AUIs by guiding the definition of adaptation rules with *What-When-How* questions. These are scaled and adapted to the dimensions of adaptation features, adaptation targets, and the initiator of adaptation.

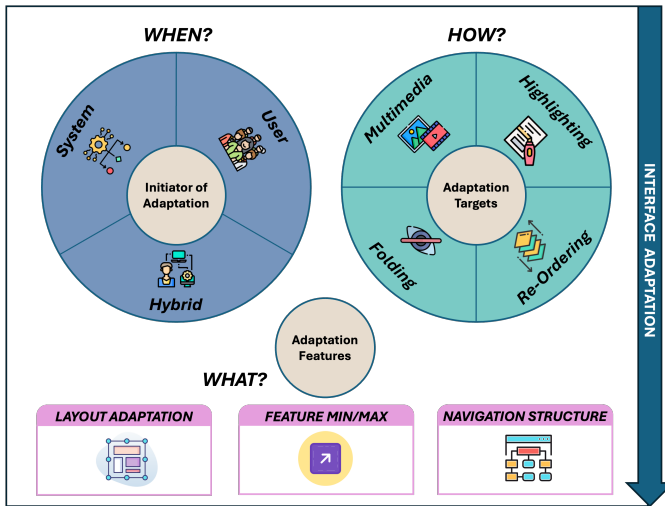


Fig. 3. Framework for the design of AUIs

For **initiators of adaptation**, create a taxonomy that categorizes triggers along three dimensions:

- *User-driven initiators*: require explicit characterization of user actions that should trigger adaptations. Document these through mappings of user actions, detailing interactions (e.g., button presses, gestures) that signal adaptation

needs. Establish thresholds for repeated behaviors that might indicate user preferences or difficulties.

- *System-driven initiators*: need comprehensive environmental sensing capabilities. Define contextual conditions like workstation configurations, ambient lighting changes, noise levels, and production variances that should trigger adaptations. Develop a context-aware sensing framework that monitors these conditions in real-time.
- *Hybrid initiators*: design a multimodal fusion mechanism that combines explicit user actions with implicit behavioral cues. Document physiological indicators (e.g., gaze patterns, interaction hesitation) and contextual factors that together signal adaptation needs.

For **adaptation targets**, conduct card sorting exercises with manufacturing workers to understand their mental models of interface organization. Use these insights to define specific adaptation mechanisms:

- *Content reordering*: establish priority algorithms based on task relevance, frequency of use, and user expertise levels. Document rules for when and how interface elements should dynamically reposition.
- *Highlighting mechanisms*: define visual treatment guidelines (color, size, animation) and determine intensity scaling based on urgency and importance.
- *Information folding*: establish information hierarchy principles that determine which content to expand or collapse. Define criteria for hiding secondary information and how users can re-access it.
- *Multimedia enhancements*: document rules to determine when to supplement text with images, videos, or audio guidance, based on task complexity and user preferences.

For **adaptation features**, combine the initiator of adaptation and the adaptation target to identify how this interaction reflects on the interface. The adaptation features can be classified as one of the following:

- *Layout adaptation*: modifies visual elements, including text font and size, color schemes, spacing, and content arrangement. Techniques like folding or reordering elements can enhance readability and visual relevance.
- *Task feature minimization/maximization*: adjusts the amount and complexity of information presented based on the current context. For instance, exploit visual folding to hide non-key content.

- *Navigation structure*: modifies navigation flows and links to reflect user needs or specific contexts. Adaptation targets, such as highlighting, reordering, and visual reduction, can optimize interface navigation.

A formal adaptive interface requirements document should be derived from this analysis, yielding:

- A classification of adaptation triggers based on observed interaction patterns.
- Prioritized adaptation targets that address identified usability challenges.
- Empirically-grounded adaptation mechanisms aligned with user preferences.

The process of identifying the interface specification logics should follow an iterative methodology: starting from a minimum viable product implementing core adaptation features, subsequent iterations will progressively introduce more sophisticated adaptation mechanisms.

5) *Adaptation logics development and integration*: This step covers the development of a logical framework for adaptation decision-making. The developer can use a hybrid approach combining rule-based systems for immediate implementation and AI components for more sophisticated behaviors or long-term improvement (e.g., Reinforcement Learning (RL) and recommendation systems). For rule-based algorithms, the developer maps adaptation specifications to formal if-then rules, using a structured rule language and implementing priority mechanisms to resolve conflicts when multiple rules apply simultaneously. Then, theoretical adaptation rules are implemented as functional software components through several interconnected activities. A logging system is desirable to capture all adaptation decisions.

6) *AUIs testing and evaluation*: This step requires to design an evaluation framework to compare the AUI with the static UI. For a fair comparison, participants must represent the same user demographics as in step 2, ensuring a mix of experience levels. Also, the same testing environment should be adopted, collecting the same initial metrics to check for improvements, integrating also new ones focused on the adaptation validation and specific to the AUI characteristics. The test should include routine operations and unusual situations that trigger adaptive behaviors. Table III categorizes some evaluation metrics, which can be customized or extended depending on the application.

7) *Review, refinement, and iteration*: This phase transforms evaluation results into concrete improvements through several structured activities. A refined adaptation model must be developed based on evaluation insights. This refinement must address triggering threshold adjustments for specific adaptations, visual implementation enhancements, timing or progression modifications of adaptation sequences, and improved transparency of adaptation logics for users.

Guidelines for future adaptive interface development must be established based on project learnings, encompassing best practices for defining adaptation rules, interface design patterns supporting effective adaptation, implementation archi-

TABLE III
EVALUATION METRICS FOR AUIs

Category	Metric	Description
Performance Metrics	Completion time	Measures the interface efficiency with step completion times.
	Number of interactions	Tracks all user interactions, including unnecessary ones, to assess information clarity.
	Error rate	Counts process mistakes, that reflect usability and cognitive load.
	Navigation efficiency	Counts backtracking, repetitions or detours within the interface.
User Interaction Metrics	Number of suggestion requests	Evaluates how often users request additional information.
	Preferred adaptation mode	Analyzes which adaptation mechanism is most frequently accepted.
	Interaction patterns	Captures user preferences in navigating different interface elements.
Cognitive Metrics	Workload assessment	Quantifies perceived task difficulty (e.g., NASA-TLX).
	Gaze tracking	Identifies areas of user focus and cognitive demand.
	Completion confidence rating	Measures how confident users feel during task execution.
Subjective User Feedback	Post-test usability questionnaires	Collects user perceptions of ease of use and adaptability (e.g., System Usability Scale (SUS), User Experience Questionnaire (UEQ)).
	Preference ranking	Allows users to rate their experience with the adaptation modes.
	Qualitative feedback	Gathers insights on frustration points, feature usefulness, and improvement suggestions.

tectures balancing flexibility with performance, and evaluation methodologies specific to adaptive interfaces.

A continuous improvement process must be established where usage data from deployed interfaces feeds back into adaptation algorithm refinement, ensuring ongoing enhancement of the adaptive system beyond initial implementation.

IV. PROTOCOL APPLICATION

To validate the protocol introduced in section III, it has been applied in a simplified yet representative experimental manufacturing scenario. The experimental application focuses on assembling a LEGO® forklift following a 16-step assembly process requiring the execution of 3 operations typical of manufacturing: picking, assembly, and quality control. The experimental setup, depicted in Fig. 4, features a tablet showing assembly instructions, and a dedicated storage box containing the required LEGO® components.

A. Static interface design

Following the first step of the protocol, a static interface (Fig. 5) was designed with a focus on simplicity and functionality, featuring interactive buttons to enhance usability. It includes a title displaying the current task or activity, navigation buttons to move back and forth through the assembly process, and a progress bar showing the process advancement.

Users can access five types of instructional content via dedicated buttons: i) *short text*, offering quick and concise

guidance; ii) *long text*, providing detailed and comprehensive instructions; iii) *single components image*, illustrating the individual parts involved; iv) *assembly image*, showing the assembled components; and v) *video*, presenting a step-by-step demonstration of the operation.

As shown in Fig. 5, some instructional content might be hidden or unavailable, depending on the current operation (i.e., picking, assembly, quality control).

B. Static interface testing

Ten volunteer users, aged between 23 and 40, were involved in the experiment. All of them agreed to participate through a consent form, and data were anonymized.

Participants were tasked with assembling the LEGO® fork-lift, and before starting the activity, they provided demographic and background information, including gender, age range, and previous experience with LEGO®.

At the beginning of each step, all instructional content is hidden from view. Users must interact with specific buttons to reveal the relevant information before proceeding with the picking/assembly/quality control operation. This approach ensures that every user interaction is recorded in an interaction log, which is going to be analyzed in future steps.

The experiment was supervised by an expert who monitored the assembly process throughout. If a participant proceeded to the next step without correctly completing the previous one, the expert advised them to press the previous button, review the instructions, and correct the step before continuing.

A total of 625 interactions were logged during the study, used for deeper analysis and exploration of potential correlations between user characteristics and interaction patterns or task performance. Furthermore, post-task interviews with the participants allowed to understand the critical points of the interface and what they was expecting step by step.

C. Interaction data and user feedback analysis

The interaction logs were analyzed to extract key insights, including completion times and button interaction sequences. On average, the total experiment duration was 472.6 seconds.



Fig. 4. The experimental setup to validate the protocol

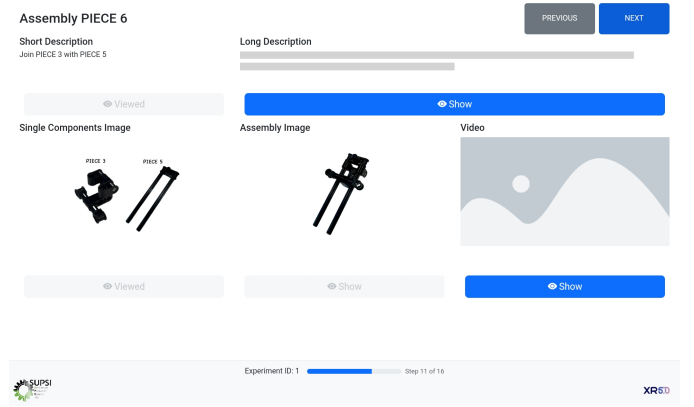


Fig. 5. The static interface guiding participants through the assembly process

Completion times varied by task type, with assembly actions taking approximately 34 seconds per action, component picking 23 seconds, and final verification requiring 42 seconds per action. Notably, the first assembly operation (step 2) took longer (65.0 seconds on average) than subsequent parts, as users started getting familiar with the UI and assembly process. By the second picking onward, users became faster and maintained stable interaction times, indicating familiarity with the interface. A similar pattern emerged for picking times, with the first picking lasting longer. Users progressively reduced assembly times until part 6, where a significant increase was observed due to the complexity of the operation. This step often required video assistance, reinforcing that users relied on videos only in moments of high difficulty.

Analysis of information suggestion requests revealed significantly different behaviors among participants and steps, as depicted in Fig. 6. Short texts were the most frequently used resources, suggesting a strong preference for quick, text-based instructions. Visual content, including single component images and assembly-focused content, played a secondary but relevant role. At the same time, videos and long texts were the least preferred, as highlighted in Table IV.

A key distinction emerged between expert (i.e., participants who were among the 50% faster ones) and novice users (i.e., participants who were among the 50% slower ones). Experts preferred visual content (31.7% for individual parts, 27.3% for assembly images), while novices relied more on text instructions (59.5% short text, 16.3% long text). This suggests that experts prefer direct visual guidance, while

TABLE IV
REQUESTED INFORMATION SUGGESTIONS

Information	Suggestion requests	Percentage
Short text	101	35.7%
Long text	31	11.0%
Single image	55	19.4%
Assembly image	70	24.7%
Video	26	9.2%

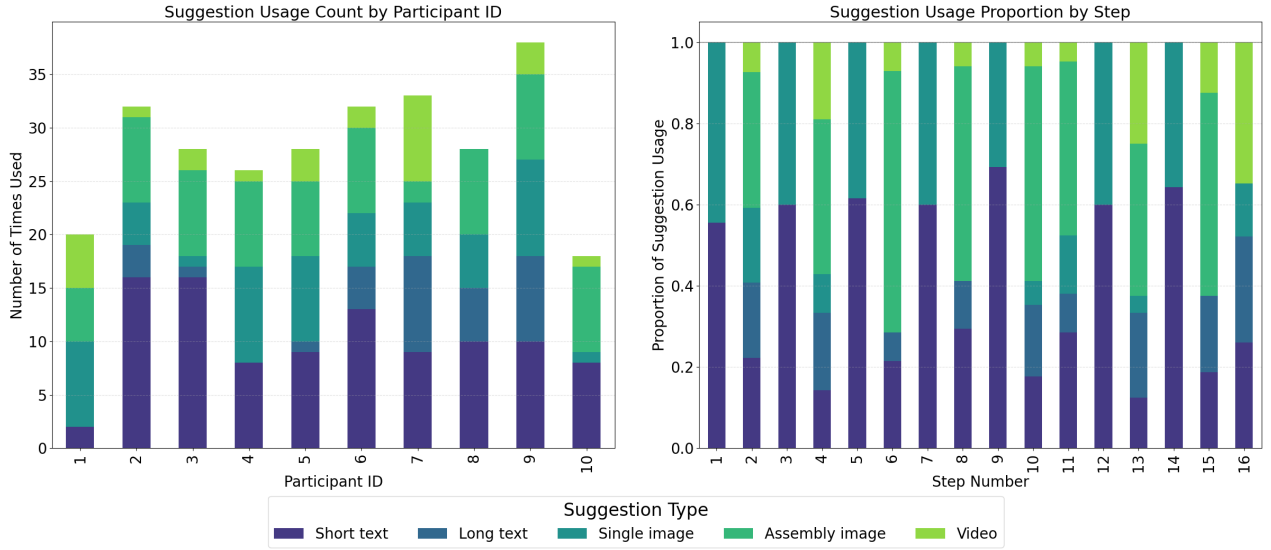


Fig. 6. Information suggestion requests per participant and step

novices require more descriptive textual support.

Moreover, transition patterns showed a strong reliance on short texts and assembly views, with frequent transitions from assembly image to short description (53 occurrences), from short description to single image (39 occurrences), and from single image to assembly image (30 occurrences).

These insights highlight opportunities to enhance guidance mechanisms through AUI integration to improve UX, accelerate assembly time, and reduce errors. This requires refining content presentation by prioritizing concise descriptions and images. Additionally, the interface should dynamically adjust information suggestions based on the assembly phase and user behavior, ensuring more relevant and efficient guidance.

D. Adaptation logics specification

The adaptation logic is designed to trigger changes in the interface based on specific events within the application. It defines which elements should be modified and when, following the Event-Condition-Action architecture, commonly used in AUI [10]. This structure integrates with events generated during the construction of the LEGO® forklift and the application’s specifications. For the adaptive logic specification, the adaptation focuses on *task feature minimization*, ensuring that users are presented with only the most relevant information for each task. This *folding* adaptation strategy simplifies the interface by emphasizing essential elements. This was implemented through both *system-driven* and *user-driven* initiators. The following set of adaptation rules was to focus on managing the complexity and quantity of displayed information:

- The system displays information adapting to the user’s specific preferences, based on their past interaction patterns. For example:
 - Event: a new assembly step is initiated.

- Condition: in past interactions, the user frequently requested training videos.
- Action: the interface displays a training video while hiding less frequently used instructions.

- The system displays information based on the ‘optimal path’ determined through data analysis, aligning closely with the preferences and behaviors observed in the test sample. For example:

- Event: a new picking step is initiated.
- Condition: data analysis revealed that users prefer image-based instructions for this type of step.
- Action: the interface displays image-based instructions while hiding the other types.

For the initial testing of these adaptive logics, only these few specifications were utilized because the focus was on validating the effectiveness of the protocol itself, ensuring the interface adapts appropriately. Factors such as user skills, task completion times, or other contextual variables were not incorporated at this stage. This decision was made to simplify the testing process and isolate the impact of the implemented adaptation strategies, allowing for a clearer evaluation of their functionality and user acceptance. Future iterations will consider additional parameters to further refine and personalize the adaptive logic.

E. Adaptation logics development and integration

To implement AUIs, different strategies were developed following the adaptation logic specification. Two simple yet effective adaptation modes were defined alongside a static baseline to enable structured comparison. Each mode starts with a single unlocked information per each step, allowing users to request additional suggestions via corresponding buttons, thereby indicating their preferences. The adaptation modes are defined as follows:

- **Pattern-guided Mode** determines the initial information displayed based on common interaction patterns observed during the static interface testing phase. For example, it may automatically present a component image for picking tasks or an assembly image for assembly tasks.
- **Preference-adaptive Mode** begins by showing the same content as the Pattern-Guided Mode. However, it continuously adapts by tracking which information users request most frequently for each task category (picking, assembly, control). Recent user selections are given higher importance to reflect evolving preferences and guide information pre-selection in subsequent steps.

Within these modes, if a user proceeds to the next step without requesting further suggestions, the initially provided information is considered sufficient, even if it may not be the preferred option. Conversely, when additional suggestions are requested, those selections are likely more informative and essential for the user. Sometimes, a combination of specific suggestions may be necessary to complete a task effectively, as a single piece of information might not always be sufficient.

F. AUIs testing and evaluation

To compare the AUI with a static UI, a **Static Mode** was developed and used as a baseline condition. In this mode, the *short text*, identified as the most frequently selected content in the static interface testing, is always visible by default. Other types of content can be accessed by interacting with the corresponding buttons.

A total of 30 participants were involved in the validation step. Each participant interacted with only one type of interface. The defined modes are integrated into the UI to adjust the level of guidance provided dynamically. Independently of the mode, the user journey follows a structured approach:

- The execution starts with a single predefined information for each step, as defined in Table V.
- Users can request additional suggestions through interface interactions.
- Once a specific step is completed, the system selects future suggestions based on the adaptation logics.
- User interactions, information requests and errors are tracked to compute evaluation metrics.

This approach persists until the assembly procedure concludes. Each participant completes a single assembly task using a designated adaptation logic. To ensure fair evaluation, 30 new participants—distinct from the initial static

TABLE V
PREDEFINED CONTENT INFORMATION

Step type	Static Mode	Pattern-guided Mode	Preference-adaptive Mode
Picking	Short text	Short text or single image	Single image*
Assembly	Short text	Assembly image	Assembly image*
Control	Short text	Video	Video*

*Dynamically adapted based on user interactions

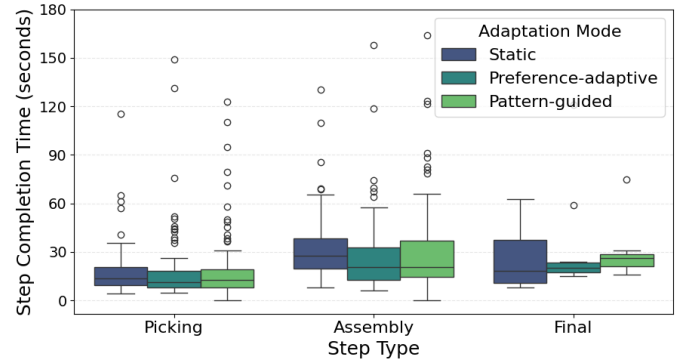


Fig. 7. Step Completion Time by Step Type and Adaptation Mode

testing—were involved in the experiments, with 10 assigned to each adaptation mode.

The following quantitative and qualitative metrics were used to assess the AUIs, defined through Table III:

- **Completion time:** measures efficiency by assessing the total duration required to complete the assembly process, reflecting cognitive load and interface usability.
- **Number of interactions and suggestion requests:** tracks how often users interact with the UI and ask for additional information suggestion besides the initial one, indicating potential gaps in information clarity or adaptability logic.
- **Error rate:** errors suggest insufficient or misleading information. They are manually annotated by the experiment supervisor based on assembly errors.
- **Qualitative metrics questionnaires:** collects subjective feedback on UI's adaptability, usefulness, clarity, usability and users satisfaction using targeted questions.

The validation metrics' evaluation led to several observations. The step completion times across different adaptation modes and step types are quite similar, with the Preference-adaptive mode being the fastest and the Static mode slightly slower than the Pattern-guided mode, as shown in Fig. 7.

Fig. 8 presents quantitative and qualitative metrics. The Static mode exhibited a higher number of interactions and suggestion requests, indicating potential confusion and misalignment with user needs. In contrast, both adaptive modes performed better, requiring fewer additional suggestions. The number of errors is significantly lower in the adaptive modes, likely due to improved UI clarity and suggestions. Moreover, the qualitative metrics demonstrated better performance in the adaptive modes, especially in terms of usefulness, clarity, usability, and user satisfaction, with the Pattern-guided mode being the best. Interestingly, some participants reported a sufficient level of adaptation even with the Static UI.

G. Review, refinement, and iteration

Following the evaluation in the previous phase, findings were analyzed to refine the adaptation strategies. The primary focus is on optimizing information presentation to minimize unnecessary interactions while ensuring sufficient guidance. Progressive disclosure should be structured to provide short

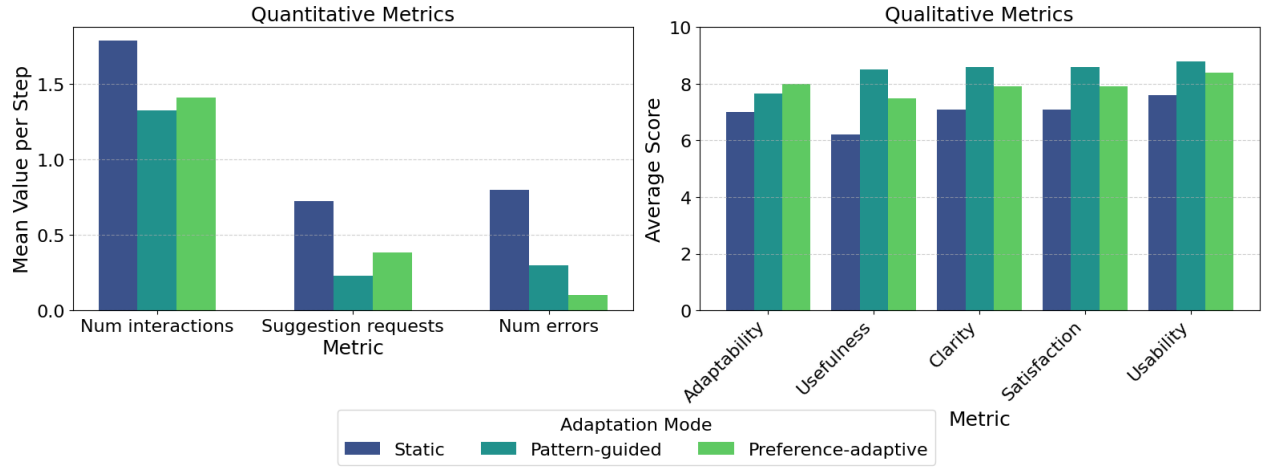


Fig. 8. Comparison of quantitative and qualitative metrics by adaptation mode

text initially, followed by assembly views, with video reserved for complex steps. Videos should be designed as problem-solving resources rather than primary guidance. Step-specific resource emphasis should prioritize short text and single-piece views early on, assembly views in the middle, and a mix of short text and assembly views for later steps.

Future work should focus on A/B testing of personalized interfaces. Moreover, extensive data collection would enable the development of ML models or recommendation systems targeted at providing information suggestions based on user preferences, characteristics and past interactions. Additional areas for refinement include introducing user clustering to identify participants with similar behaviors and preferences, evaluating a combination of unlocked information based on execution patterns and conducting additional iterations with other participants to validate existing findings across different user profiles. Extensions will also incorporate metrics such as user skills and step completion times into the interface adaptation process. These factors will provide deeper insights into individual performance, enabling more accurate identification of each user's cluster and, consequently, allowing the system logic to better adapt to the user's specific situation. Based on these considerations, iteration of protocol phases enables continuous improvement of the AUI, ensuring its effectiveness in supporting task execution and minimizing user effort.

V. CONCLUSION

The advancement of HMI through adaptive interfaces in manufacturing environments has shown significant potential to improve productivity and efficiency. Key findings indicate that adaptive interfaces reduce assembly time variability and errors, while enhancing user satisfaction by personalizing interactions based on user behavior. The implementation of adaptive algorithms has proven instrumental in optimizing these interfaces, allowing for a more intuitive and responsive manufacturing ecosystem.

However, the main limitation of this work lies in the evaluation of the protocol, which is based on a single laboratory-built

use case. Further validation is needed with diverse applications and more complex adaptation methods. Furthermore, the assembly processes were not tracked, allowing participants to advance in the process without having correctly completed the previous steps. Future work includes the installation of a camera that monitors the task progress and prevent users to advance in case of errors or missing operations. Also, more features and targets will be considered to improve the adaptive and user-centered design of the AUI.

Next steps to improve the use case include: i) refining the developed protocol by improving the performance of the system in the terms of response-time; ii) applying the presented protocol to other case studies, developing further scenarios; iii) improving the system by including usability aspects reported during the user study and 'information' quality (e.g., many users complained about the difficulty of understanding the positions of the different pieces in an image, which was one of the main cause of error, caused by the similar colors of the LEGO® pieces used in the experiment); iv) refining the specification of the adaptive logics by also taking into consideration human factors such as skill and mental fatigue.

MATCH & CONTRIBUTION

This work proposes a systematic approach providing structured assistance and guidance in the design of AUIs. By addressing adaptability, usability, and user trust, the proposed framework bridges the gap between theoretical research and practical application. In this view, the study aligns with the IEEE Technology and Engineering Management Society (TEMS) research objectives by focusing on the management and strategic implementation of emerging technology, also contributing directly to engineering management practices and technological advancements.

ACKNOWLEDGMENT

This work has received funding from the Swiss State Secretariat for Education, Research and Innovation (SERI) and has been partly supported by the European Union's research

and innovation programme under project XR5.0 (Grant n. 101135209).

REFERENCES

- [1] A. Bettoni, D. Matteri, E. Montini, B. Gładysz, and E. Carpanzano, "An ai adoption model for smes: A conceptual framework," *IFAC-PapersOnLine*, vol. 54, no. 1, pp. 702–708, 2021.
- [2] M. Da Bormida in Cugurra, S. Koussouris, V. Cutrona, and L. Canetta, "A human rights-driven ai assessment of ethical industry 5.0 transition and sustainability in circular value chains," in *2024 IEEE International Conference on Engineering, Technology, and Innovation (ICE/ITMC)*, 2024, pp. 1–9.
- [3] D. A. d. J. Pacheco and B. Iwaszczenko, "Unravelling human-centric tensions towards industry 5.0: Literature review, resolution strategies and research agenda," *Digital Business*, p. 100090, 2024.
- [4] S. Masiero, J. Qosaj, A. Bettoni, and B. Gladysz, "Technology-driven measures for human centricity in the manufacturing sector," in *International Association for the Management of Technology Conference*. Springer, 2024, pp. 81–88.
- [5] V. Villani, L. Sabattini, P. Barańska, E. Callegati, J. N. Czerniak, A. Debbache, M. Fahimipirehgalin, A. Gallasch, F. Loch, R. Maida *et al.*, "The inclusive system: A general framework for adaptive industrial automation," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 4, pp. 1969–1982, 2020.
- [6] E. Montini, G. Ploner, D. Matteri, V. Cutrona, P. Rocco, A. Bettoni, and P. Pedrazzoli, "Impact of collaborative robots on human trust, anxiety, and workload: Experiment findings," in *IFIP International Conference on Advances in Production Management Systems*. Springer, 2024, pp. 401–415.
- [7] D. Lindlbauer, A. M. Feit, and O. Hilliges, "Context-aware online adaptation of mixed reality interfaces," in *Proceedings of the 32nd annual ACM symposium on user interface software and technology*, 2019, pp. 147–160.
- [8] J. Y. Chew, M. Kawamoto, T. Okuma, E. Yoshida, and N. Kato, "Adaptive attention-based human machine interface system for teleoperation of industrial vehicle," *Scientific reports*, vol. 11, no. 1, p. 17284, 2021.
- [9] D. Reguera-Bakhache, I. Garitano, C. Cernuda, R. Uribeetxeberria, U. Zurutuza, and G. Lasas, "An adaptive industrial human-machine interface to optimise operators working performance," in *2021 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*. IEEE, 2021, pp. 1213–1219.
- [10] G. Ghiani, M. Manca, F. Paternò, J. Rett, and A. Vaibhav, "Adaptive multimodal web user interfaces for smart work environments," *Journal of Ambient Intelligence and Smart Environments*, vol. 7, no. 6, pp. 701–717, 2015.
- [11] H. Oestreich, M. Heinz-Jakobs, P. Sehr, and S. Wrede, "Human-centered adaptive assistance systems for the shop floor," in *Human-Technology Interaction: Shaping the Future of Industrial User Interfaces*. Springer, 2022, pp. 83–125.
- [12] V. N. S. K. Challa, "Adaptive user interfaces: Personalizing user experience user profiling and modeling," *Journal of Artificial Intelligence & Cloud Computing*, vol. 2, no. 2, pp. 1–12, 2023.
- [13] S. Dell'Oca, D. Matteri, E. Montini, V. Cutrona, Z. M. Barut, and A. Bettoni, "Improving collaborative robotics: Insights on the impact of human intention prediction," in *Human-Friendly Robotics 2024*. Cham: Springer Nature Switzerland, 2025, pp. 1–15.
- [14] A. Ghosh, "Enhancing human-computer interaction in smart manufacturing environments: Designing intuitive and contextual interfaces for operators and supervisors," *International Research Journal of Modernization in Engineering Technology and Science*, vol. 4, no. 12, pp. 2151–2155, 2022.
- [15] A. Carrera-Rivera, F. Larrinaga, G. Lasas, G. Martinez-Arellano, and G. Unamuno, "Adaptui: a framework for the development of adaptive user interfaces in smart product-service systems," *User Modeling and User-Adapted Interaction*, pp. 1–52, 2024.
- [16] F. Hartwich, C. Hollander, D. Johannmeyer, and J. F. Krems, "Improving passenger experience and trust in automated vehicles through user-adaptive hmis: "the more the better" does not apply to everyone," *Frontiers in Human Dynamics*, vol. 3, p. 669030, 2021.
- [17] S. Brdnic, T. Heričko, and B. Šumak, "Intelligent user interfaces and their evaluation: A systematic mapping study," *Sensors*, vol. 22, no. 5830, 2022.
- [18] X. Li, D. Zhao, and J. Zhao, "A design case study: 3d printer software interface design based on home users preferences knowledge," in *Proceedings of the Design Society: International Conference on Engineering Design*, vol. 1, no. 1. Cambridge University Press, 2019, pp. 639–648.
- [19] F. Rizzoglio, M. Casadio, D. De Santis, and F. A. Mussa-Ivaldi, "Building an adaptive interface via unsupervised tracking of latent manifolds," *Neural Networks*, vol. 137, pp. 174–187, 2021.
- [20] L. Späker, B. G. Mark, and E. Rauch, "Development of a morphological box to describe worker assistance systems in manufacturing," *Procedia Manufacturing*, vol. 55, pp. 168–175, 2021.
- [21] G. Fenza, V. Loia, and G. Nota, "Patterns for visual management in industry 4.0," *Sensors*, vol. 21, no. 19, p. 6440, 2021.
- [22] M. T. Moghaddam, H. Muccini, J. Dugdale, and M. B. Kjægaard, "Designing internet of behaviors systems," in *2022 IEEE 19th International Conference on Software Architecture (ICSA)*. IEEE, 2022, pp. 124–134.
- [23] M. Ciccarelli, A. Brunzini, A. Papetti, and M. Germani, "Interface and interaction design principles for mixed reality applications: the case of operator training in wire harness activities," *Procedia Computer Science*, vol. 204, pp. 540–547, 2022.
- [24] M. Bernabei and F. Costantino, "Design a dynamic automation system to adaptively allocate functions between humans and machines," *IFAC-PapersOnLine*, vol. 56, no. 2, pp. 3528–3533, 2023.
- [25] K. Todi and T. R. Jonker, "A framework for computational design and adaptation of extended reality user interfaces," *arXiv preprint arXiv:2309.04025*, 2023.
- [26] A. Carrera-Rivera, D. Reguera-Bakhache, F. Larrinaga, G. Lasas, and I. Garitano, "Structured dataset of human-machine interactions enabling adaptive user interfaces," *Scientific Data*, vol. 10, no. 1, p. 831, 2023.
- [27] A. K. Sinha, C. Y. Choi, and D. W. Rosen, "Automated usability and user experience assessment for smart products," in *Proceedings of the 31st ISTE International Conference on Transdisciplinary Engineering*, ser. Advances in Transdisciplinary Engineering. IOS Press, 2024, pp. 432–441.
- [28] F. Loch, M. Fahimipirehgalin, J. N. Czerniak, A. Mertens, V. Villani, L. Sabattini, C. Fantuzzi, and B. Vogel-Heuser, "An adaptive virtual training system based on universal design," *IFAC-PapersOnLine*, vol. 51, no. 34, pp. 335–340, 2019.
- [29] T. K. Khan, P. Schworm, M. F. Glatt, C. Ugoji, A. Ebert, and J. C. Aurich, "Engaging human-centered design to maintain part manufacturing under reduced workforce restrictions," *Production Engineering*, vol. 18, no. 5, pp. 733–750, 2024.
- [30] L. Dodeja, P. Tambwekar, E. Hedlund-Botti, and M. Gombolay, "Towards the design of user-centric strategy recommendation systems for collaborative human-ai tasks," *International journal of human-computer studies*, vol. 184, p. 103216, 2024.
- [31] L. Gualtieri, F. Fraboni, H. Brendel, L. Pietrantoni, R. Vidoni, and P. Dallasega, "Updating design guidelines for cognitive ergonomics in human-centred collaborative robotics applications: An expert survey," *Applied ergonomics*, vol. 117, p. 104246, 2024.
- [32] K. Sallam, M. Mohamed, and A. W. Mohamed, "Internet of things (iot) in supply chain management: challenges, opportunities, and best practices," *Sustainable machine intelligence journal*, vol. 2, pp. 3–1, 2023.
- [33] V. Villani, L. Sabattini, G. Zanelli, E. Callegati, B. Bezzi, P. Barańska, Z. Mockało, D. Żołnierczyk-Zreda, J. N. Czerniak, V. Nitsch *et al.*, "A user study for the evaluation of adaptive interaction systems for industrial workplaces," *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 4, pp. 3300–3310, 2021.
- [34] D. Mourtzis, J. Angelopoulos, and V. Zogopoulos, "Integrated and adaptive air maintenance and shop-floor rescheduling," *Computers in Industry*, vol. 125, p. 103383, 2021.
- [35] G. Fabio, C. Giuditta, P. Margherita, and R. Raffaeli, "A human-centric methodology for the co-evolution of operators' skills, digital tools and user interfaces to support the operator 4.0," *Robotics and Computer-Integrated Manufacturing*, vol. 91, p. 102854, 2025.
- [36] Z. Keskin, D. Joosten, N. Klasen, M. Huber, C. Liu, B. Drescher, and R. H. Schmitt, "Llm-enhanced human-machine interaction for adaptive decision making in dynamic manufacturing process environments," *IEEE Access*, 2025.
- [37] M. H. Miraz, M. Ali, and P. S. Excell, "Adaptive user interfaces and universal usability through plasticity of user interface design," *Computer Science Review*, vol. 40, p. 100363, 2021.